

Predictive Maintenance using Machine Learning (NASA CMAPSS Dataset)

Internal Working Document (Not Final Report)

Note: This document is an internal working file created for understanding, planning, and collaboration among team members. It is not the final project report.

The purpose of this document is to break down the project into clear steps, track progress, and maintain a structured approach while building the predictive maintenance model.

Dataset Source: [NASA CMAPSS \(Commercial Modular Aero-Propulsion System Simulation\)](#)

1. Introduction

In modern industries, unexpected machine failures can lead to high costs, downtime, and safety risks. To reduce such issues, predictive maintenance techniques are used to estimate when a machine is likely to fail so that maintenance can be performed in advance.

In this project, we aim to build a machine learning model that predicts the Remaining Useful Life (RUL) of a machine. RUL represents how many operational cycles are left before the machine fails.

We are using the NASA CMAPSS dataset, which contains simulated data of aircraft jet engines. Each engine is monitored over time using multiple sensors, and the data shows how the engine degrades until failure.

The goal of this project is to understand the dataset, process the data correctly, and build a model that can predict machine failure in advance using sensor readings.

2. Objective

The main objective of this project is to develop a predictive maintenance system using machine learning.

The model aims to predict the Remaining Useful Life (RUL) of a machine based on sensor data.

Based on the predicted RUL:

- If RUL is low → maintenance should be performed
- If RUL is high → machine can continue operating safely

The project focuses on understanding how data-driven techniques can be used to reduce unexpected machine failures.

3. Dataset Overview

We are using the NASA CMAPSS (Commercial Modular Aero-Propulsion System Simulation) dataset for this project.

This dataset contains simulated data of aircraft jet engines operating over time until failure.

Key characteristics of the dataset:

- Each engine is identified by a unique engine ID
- Each row represents one operational cycle of an engine
- The dataset includes:
 - 3 operational settings
 - Multiple sensor measurements
 - The data shows how engine performance degrades over time

Files used in this project:

- train_FD001.txt → contains complete engine life data (from start to failure)
- test_FD001.txt → contains incomplete engine data (stops before failure)
- RUL_FD001.txt → contains actual Remaining Useful Life values for test data

Note:

We are using only the FD001 dataset for simplicity and clarity.

Additional Details about Dataset Columns

Apart from the engine ID and cycle number, the dataset contains multiple other columns which play a crucial role in the project.

Operational Settings:

- The next 3 columns represent operational settings of the engine
- These include conditions such as altitude, speed, or environmental factors under which the engine is operating

Sensor Measurements:

- The remaining 21 columns represent sensor readings from different parts of the engine
- These sensors measure parameters such as temperature, pressure, and internal performance of the engine

Importance of these columns:

- Sensor values reflect the health and condition of the engine at each cycle
- As the engine degrades over time, these sensor values change accordingly
- These changes help the model learn patterns of degradation

Conclusion:

The machine learning model uses these sensor values (along with cycles) as input to predict the Remaining Useful Life (RUL) of the engine.

4. Key Concepts

4.1 What is Predictive Maintenance

Predictive maintenance is a technique used to determine when a machine is likely to fail so that maintenance can be performed before the failure occurs.

Instead of repairing machines after failure, predictive maintenance helps in reducing downtime and improving efficiency by predicting failures in advance.

4.2 What is RUL (Remaining Useful Life)

Remaining Useful Life (RUL) represents the number of operational cycles a machine can continue to function before it fails.

It is calculated as:

$$\text{RUL} = \text{Maximum Cycle} - \text{Current Cycle}$$

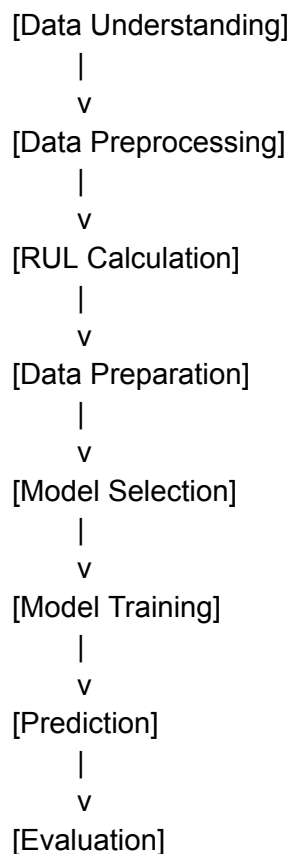
A lower RUL indicates that the machine is close to failure, while a higher RUL indicates that the machine is still in a healthy condition.

Predicting RUL is the main goal of this project.

5. Project Workflow (Flow Representation)

The project follows a structured sequence of steps to build a predictive maintenance model. Instead of detailed explanations, this section provides a high-level flow of the complete process.

Workflow (text flowchart):



This flow represents the complete pipeline from raw dataset to final model evaluation. Each of these steps is explained in detail in the next section.

6. Detailed Project Steps

This section describes the complete step-by-step process to build the predictive maintenance model. Each step explains what needs to be done and why it is important.

6.1 Step 1: Data Understanding

Goal:

To clearly understand how the dataset is structured and what each part represents.

What to do:

- Open the file train_FD001.txt
- Observe the structure of the data
- Identify how engine IDs and cycles are arranged
- Notice that:
 - Same engine ID repeats for multiple rows
 - Cycle value increases step-by-step
 - When a new engine starts, cycle resets to 1

Understanding:

- Each engine runs from cycle 1 until failure
- Each row represents one time step (cycle)
- Sensor values represent the condition of the engine at that cycle

What to conclude:

- Engines degrade over time
- Failure happens at the last cycle of each engine

6.2 Step 2: Data Preprocessing

Goal:

To clean and structure the dataset for further processing.

What to do:

- Load the dataset into a programming environment (Python recommended)
- Assign column names:
 - id → engine ID
 - cycle → time step
 - op1, op2, op3 → operational settings
 - s1 to s21 → sensor values
- Remove extra columns if present (sometimes empty columns exist)
- Check for missing or invalid data

Understanding:

- Raw data is not labeled, so we must assign meaning to each column
- Clean data is necessary for accurate model training

What to conclude:

- Dataset is now structured and ready for further steps

6.3 Step 3: RUL Calculation

Goal:

To create the target variable (what the model will predict)

What to do:

- For each engine:
 - Find the maximum cycle value (this is the failure point)
- For each row:
 - Calculate RUL using:

$$\text{RUL} = \text{Maximum Cycle} - \text{Current Cycle}$$

Example:

If an engine fails at cycle 200:

- At cycle 150 → RUL = 50
- At cycle 199 → RUL = 1
- At cycle 200 → RUL = 0

Understanding:

- RUL tells how many cycles are left before failure
- This is what the model will learn to predict

What to conclude:

- Training dataset now has input (sensor data) + output (RUL)

6.4 Step 4: Data Preparation

Goal:

To prepare the dataset in a format suitable for model training.

What to do:

- Normalize or scale sensor values to bring them into a similar range
- This helps the model learn efficiently and prevents bias toward large values
- Select relevant features:
 - Sensor values (s1 to s21)
 - Operational settings (op1, op2, op3) if useful
- Separate input and output:
 - Input (X) → sensor + operational data
 - Output (Y) → RUL

Understanding:

- Machine learning models work better when data is scaled and organized properly
- Proper input-output separation is necessary for training

What to conclude:

- Data is now ready to be used for training the model

6.5 Step 5: Model Selection

Goal:

To choose an appropriate model for predicting Remaining Useful Life (RUL)

What to do:

- Decide between:
 - Basic Machine Learning models (e.g., Linear Regression, Random Forest)
 - Deep Learning models (e.g., LSTM)
- Since the data is time-based (sequential cycles):
 - LSTM (Long Short-Term Memory) is commonly used
 - It can capture patterns over time

Understanding:

- Simple models are easier but may be less accurate
- Deep learning models can capture complex patterns in time-series data

What to conclude:

- Model type is selected based on project requirements and understanding

6.6 Step 6: Model Training

Goal:

To train the selected model using the prepared dataset

What to do:

- Feed training data (X and RUL) into the model
- Allow the model to learn patterns between sensor data and failure
- Adjust parameters (if needed) to improve performance
- Train for multiple iterations (epochs in deep learning)

Understanding:

- The model learns how engine degradation affects failure
- Better training leads to better predictions

What to conclude:

- Model is now trained and ready for testing

6.7 Step 7: Prediction

Goal:

To use the trained model to predict RUL for unseen data

What to do:

- Input test dataset (test_FD001.txt) into the trained model
- Generate predicted RUL values for each engine

Understanding:

- Test data does not include failure information
- The model estimates how many cycles remain

What to conclude:

- Predictions simulate real-world maintenance decisions

6.8 Step 8: Evaluation

Goal:

To evaluate how accurate the model predictions are

What to do:

- Compare predicted RUL values with actual RUL values from RUL_FD001.txt

- Calculate error (difference between predicted and actual values)

Understanding:

- Lower error means better model performance
- Evaluation helps in improving the model

What to conclude:

- Model performance is measured and validated

7. Observations & Learnings

- The dataset initially appears complex due to lack of column names and raw numerical format
- Understanding engine ID and cycle structure is the key to decoding the dataset
- Engines degrade gradually over time and do not fail suddenly
- Remaining Useful Life (RUL) is the most important concept in predictive maintenance
- Proper data preprocessing is essential before applying any model
- Time-based (sequential) data requires careful handling for better predictions

8. Work Distribution (Team Planning)

This section can be updated based on team discussion.

Suggested division of work:

- Member 1: Data Understanding and Preprocessing
- Member 2: RUL Calculation and Data Preparation
- Member 3: Model Building and Training
- Member 4: Evaluation, Documentation, and Presentation

Note:

All members should understand the full project for viva and discussion.

9. Future Improvements (Optional)

- Experiment with different machine learning models for better accuracy
- Use advanced deep learning models for improved prediction
- Perform feature selection to identify the most important sensors
- Improve data preprocessing and normalization techniques
- Build a simple user interface to display predictions

10. Final Notes

This document serves as a structured guide for understanding and building the predictive maintenance project.

It will be continuously updated as the project progresses, including code implementation, results, and observations.

The focus of this project is not only to build a working model but also to clearly understand each step involved in the process.